
September 17, 2026

Helmholtz-Zentrum Dresden-Rossendorf
Bautzner Landstr. 400
Dresden, 01328, Germany

Dear Prof. Beverley J. McKeon,
Chief Editor of *Physical Review Fluids*

We wish to submit an original journal article titled “Magnetic damping of standing waves in rectangular geometries” in the *Physical Review Fluids*. This study primarily aims to calculate the magnetic damping rate of a free-surface standing wave of a conducting fluid within a rectangular domain.

In summary, our contributions are the following:

- We formulate the magnetic damping rate of a surface gravity wave in a rectangular geometry filled with an electrically conducting fluid, which is subjected to a vertical magnetic field.
- Previous studies only considered i) only one single type of wave mode $(1, 0)$, ii) shallow water approximation, iii) and uniform electrical boundary conditions (either all insulating or all conducting). The present study considered arbitrary wave modes with mixed electrical boundary conditions, disregarding the shallow water approximation.
- Furthermore, we have also formulated a new expression for the Hartmann and Shercliff boundary layer in a single-layer system, and the Hartmann boundary layer formed at the interface in a two-layer system with large electrical conductivity difference (quasi one-layer system).

We have also validated a part of the theory with a simple lab-scale sloshing experiment. Lastly, we provide a supplementary material that details all the analytical derivation along with a Python module to aid the readers in reproducing and applying our theoretical results.

I confirm that this paper, has not been published elsewhere. All the authors have approved the manuscript and agree with its submission to the *Physical Review Fluids*.

Kind regards,

Pranav Hegde,

On behalf of Thomas Gundrum and Gerrit Maik Horstmann
